

Design decisions in Ingesting and analyzing data of a global retail

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The case study: global retail

Global
Chain



Walmart

Hundreds
of Locations

Thousands
of Locations

Hundreds
of Locations

SUBWAY

Thousands
of Locations

Thousands
of Locations

Hundreds
of Thousands





10,000s of stores worldwide

- **7-Eleven:** One of the largest convenience store chains in the world, with more than **83,000** locations across 19 countries. It's particularly prominent in Japan, the United States, and various parts of Asia.
- **Subway:** The fast-food chain has over **37,000** restaurants worldwide, although it has closed some stores in recent years.
- **McDonald's:** While not a traditional retail chain, McDonald's operates more than **40,000** restaurants in over 100 countries.
- **Walmart:** It operates around **10,500** stores globally, including its subsidiaries like Sam's Club and stores in international markets like Mexico and Canada.



100s-1,000,000s distinct SKUs

- **7-Eleven:** On average, a single store carries **2,500 - 3,000 SKUs**, but this number can vary depending on location and local preferences
- **Subway:** It manages **200 - 300 SKUs** across its entire menu and supply chain
- **McDonald's:** Similarly, a typical location handles **200 to 300 SKUs**.
- **Walmart:** A typical Walmart in the U.S. carries **120,000-150,000 SKUs** across various categories (groceries, electronics, clothing, and home goods). It offers **millions of SKUs** online.



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The objective



Ingest, analyze, and support decisions

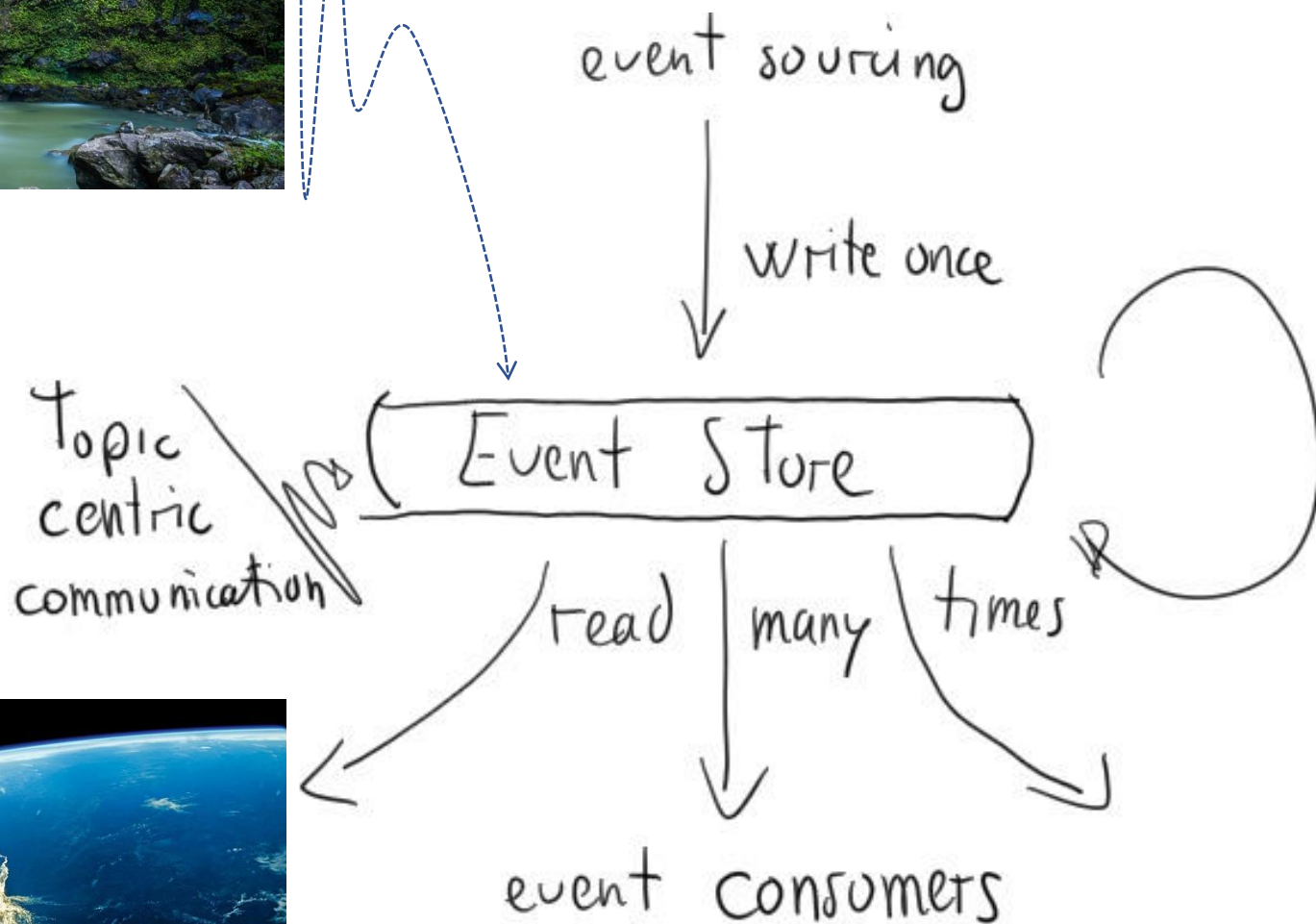
- **Ingest** sales data from a global retail, i.e., **each** single **bill** **and** **restock**
- Keep **real-time inventory** (what the **stock of each SKU** across all stores) **for** supporting supply chain decisions (e.g., **restocking** given store's **SKUs** from warehouses)
- Perform **real-time price analysis** to support store managers' decisions **to promote SKUs** (e.g., those for which there is an overstock)



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**Designing ...
(at high level)**

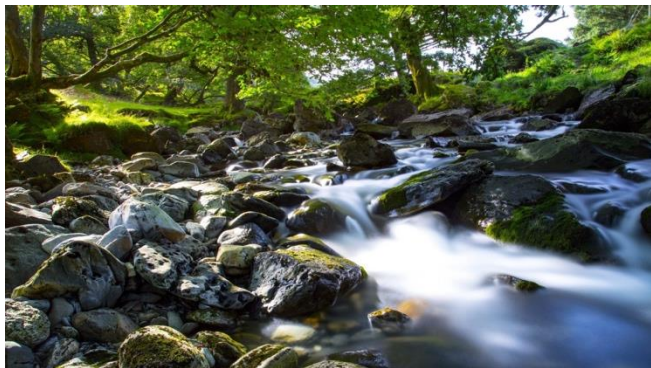
Event-based Systems



Data Stream Mng. Sys.

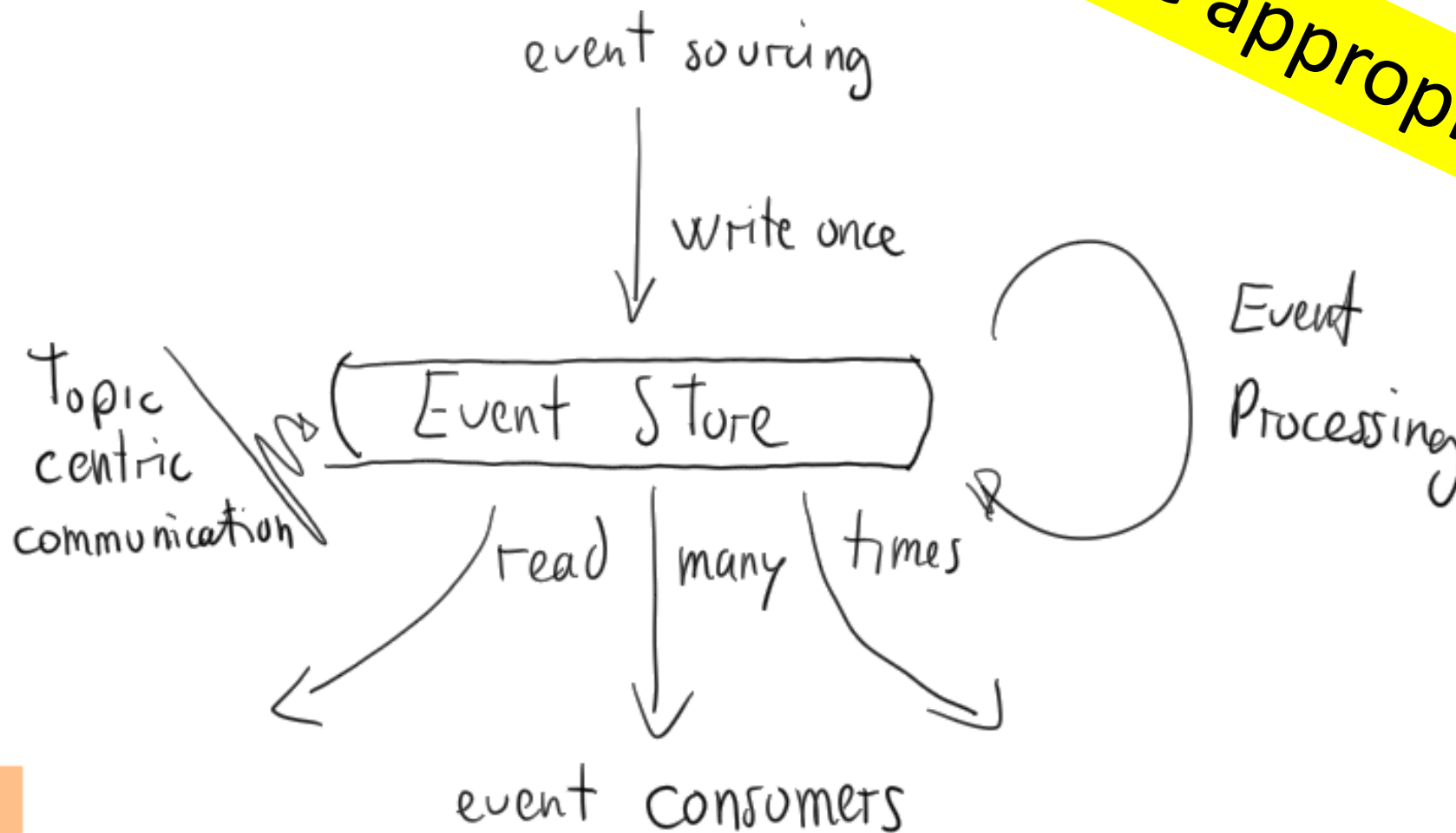


Event Processing



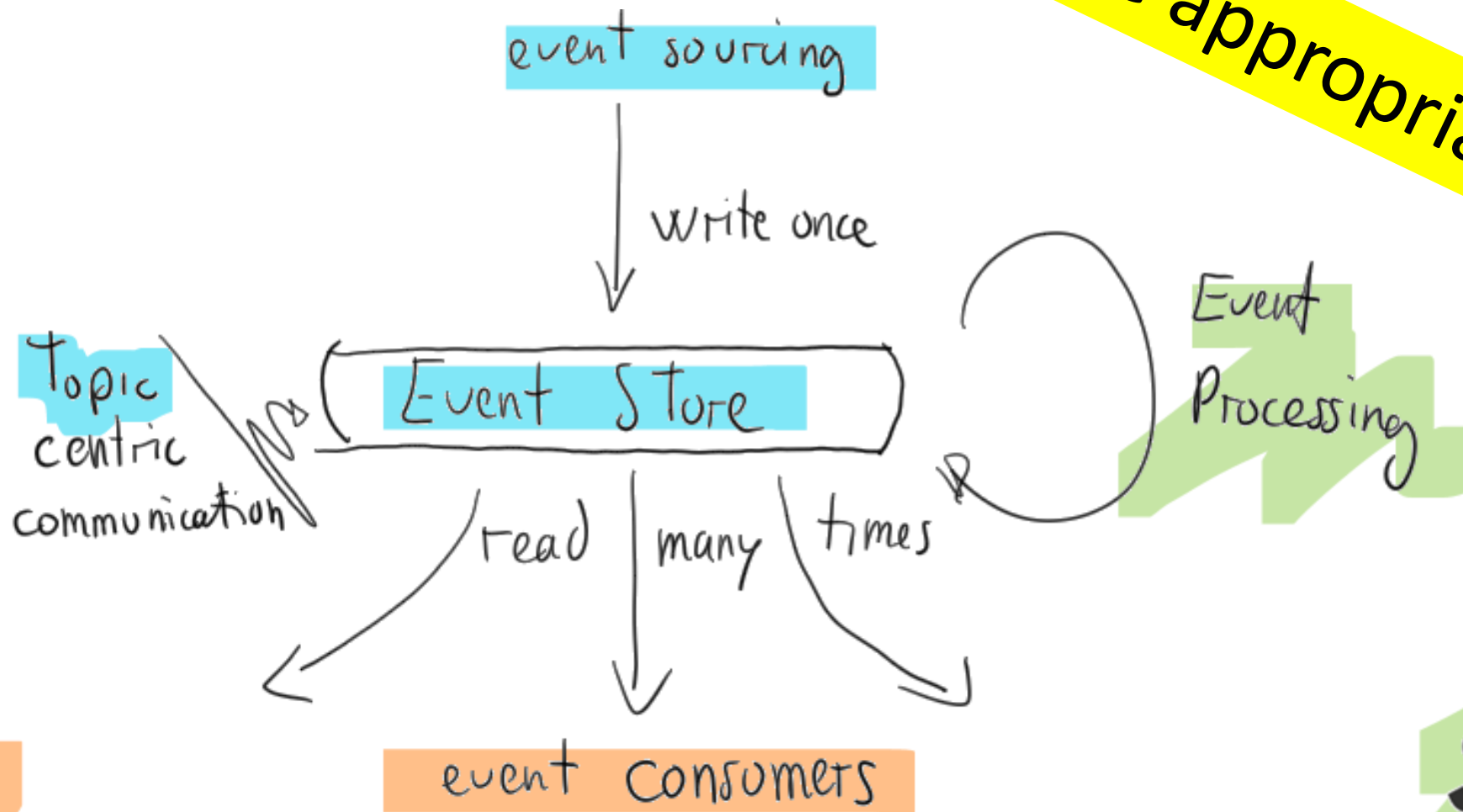
Complex Event Proc.

Color the appropriate parts



support
decisions

Color the appropriate parts



support
decisions



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Q: What are the roles of kafka?

Relevant





A: kafka ...

- ... continuously ingests the sales and restocking data
- ... stores the current inventory of each SKU for each store
- ... acts as a channel between different parts of the analytics
- ... acts as a broker between analytics results and decision support apps

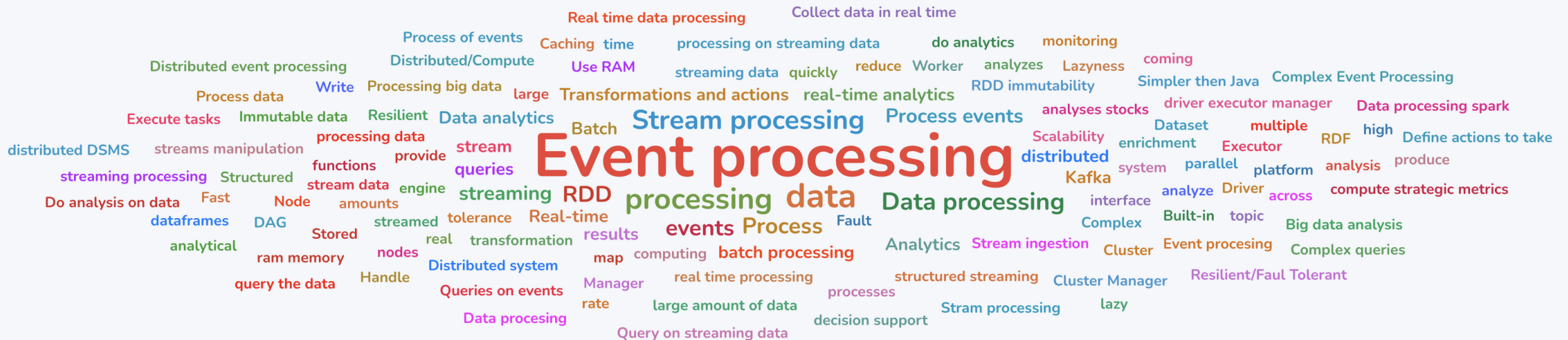


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Q: What are the roles of Spark?



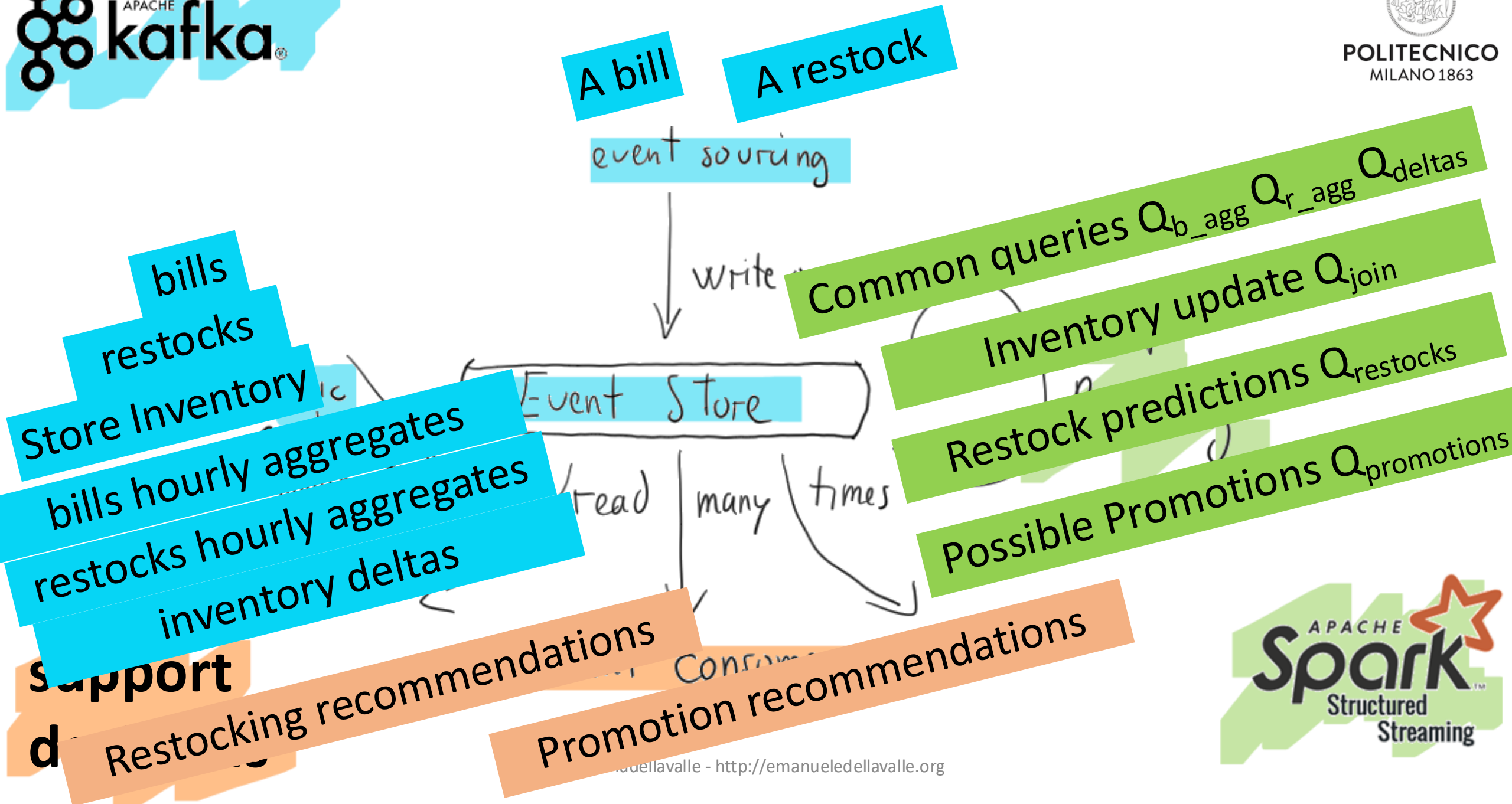
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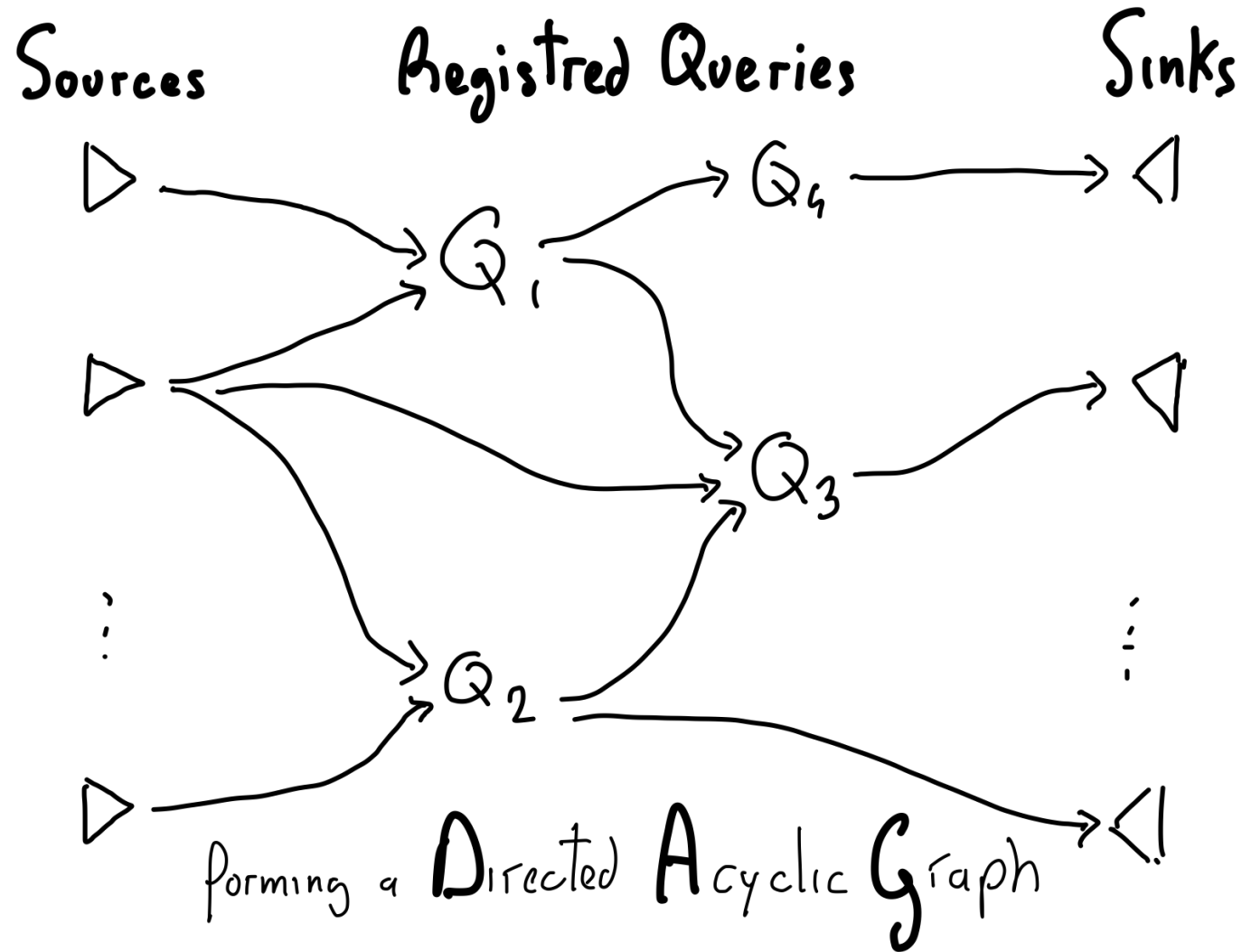


A: Spark ...

- ... aggregates bills per store, SKU, and hour (Q_{b_agg})
- ... aggregates restocks per store, SKU, and hour (Q_{r_agg})
- ... computes deltas per store, SKU, and hour (Q_{deltas})
- ... joins windowed deltas and last stocks and gets current stock (Q_{join})
- ... analyses current stocks to predict the optimal restocks ($Q_{restocks}$)
- ... analyses current stocks and prices to predict possible promotions to sale SKUs that risk overstock ($Q_{promotions}$)



Typically queries are placed in a network



Sources

bills ▷

restocks ▷

store
inventory
f-1 ▷

Registered Queries

$Q_{b.agg}$ $Q_{r.agg}$ Q_{deltas} Q_{join} $Q_{restocks}$ $Q_{promotions}$

Sinks

◁ restocking
recommendations

◁ promotions
recommendations

◁ store
inventory
f-0

Place here
the continuous queries
in a DAG



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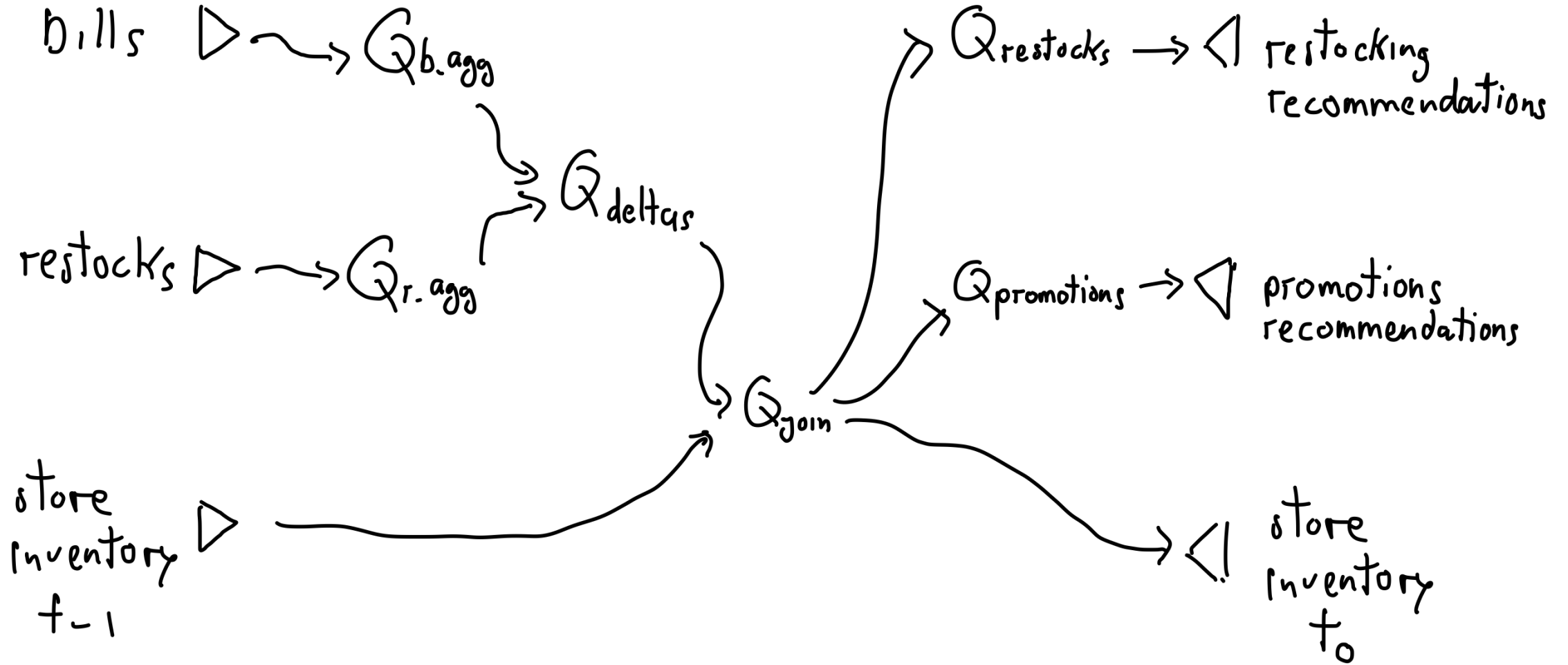
Sources

Registered Queries

Sinks

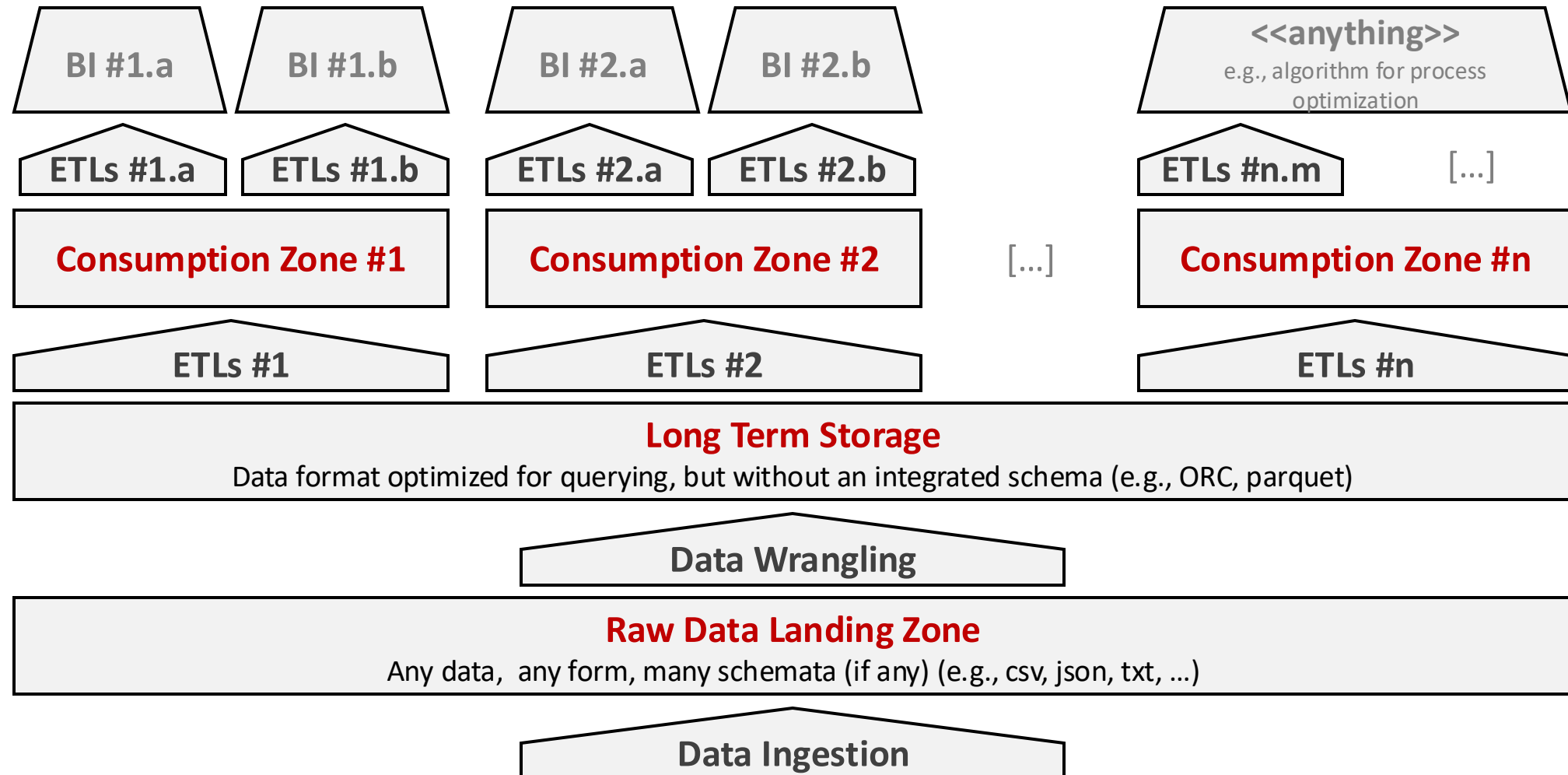


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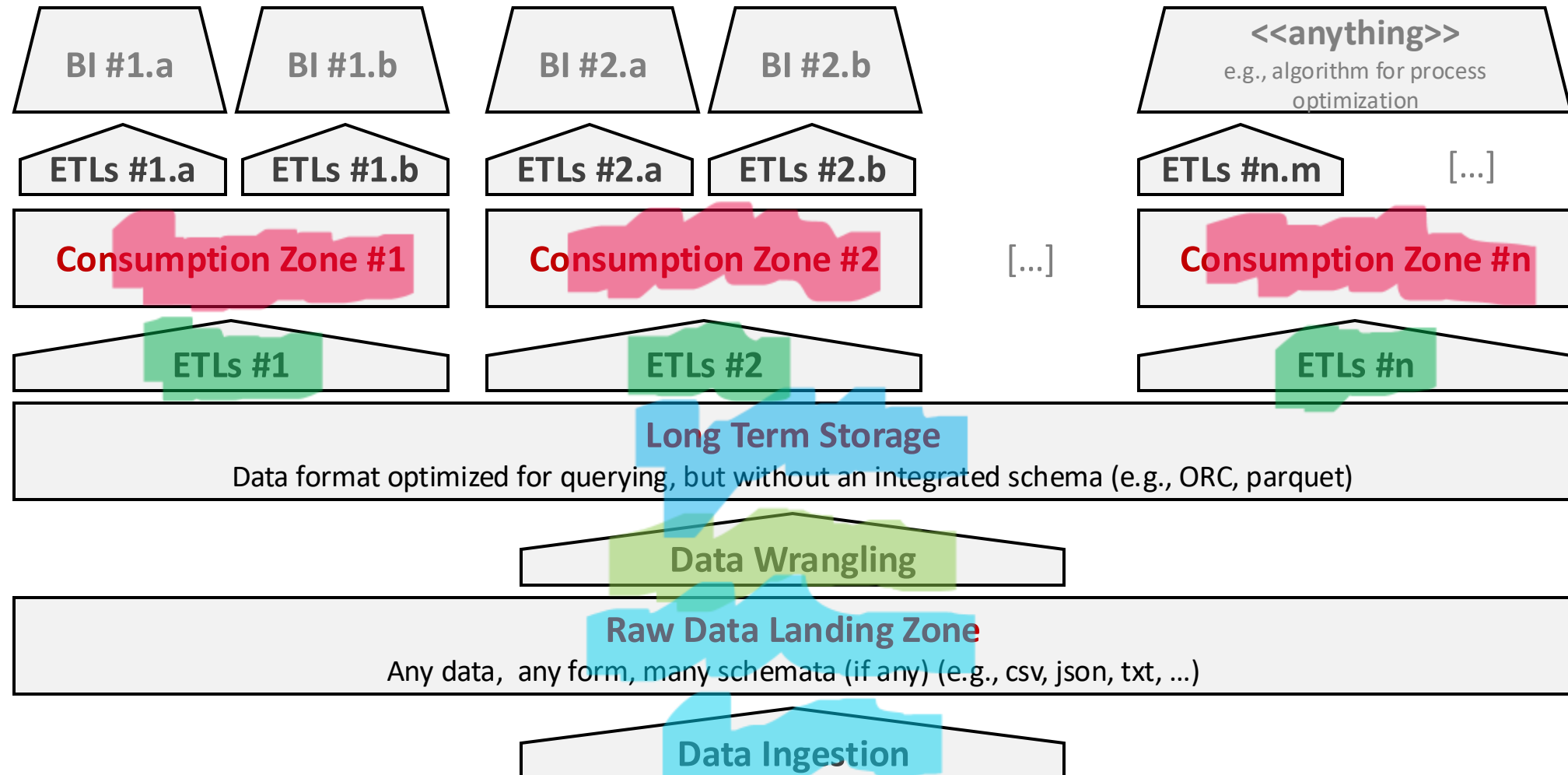




Logical Architecture of a Data Platform



Logical Architecture of a Data Platform

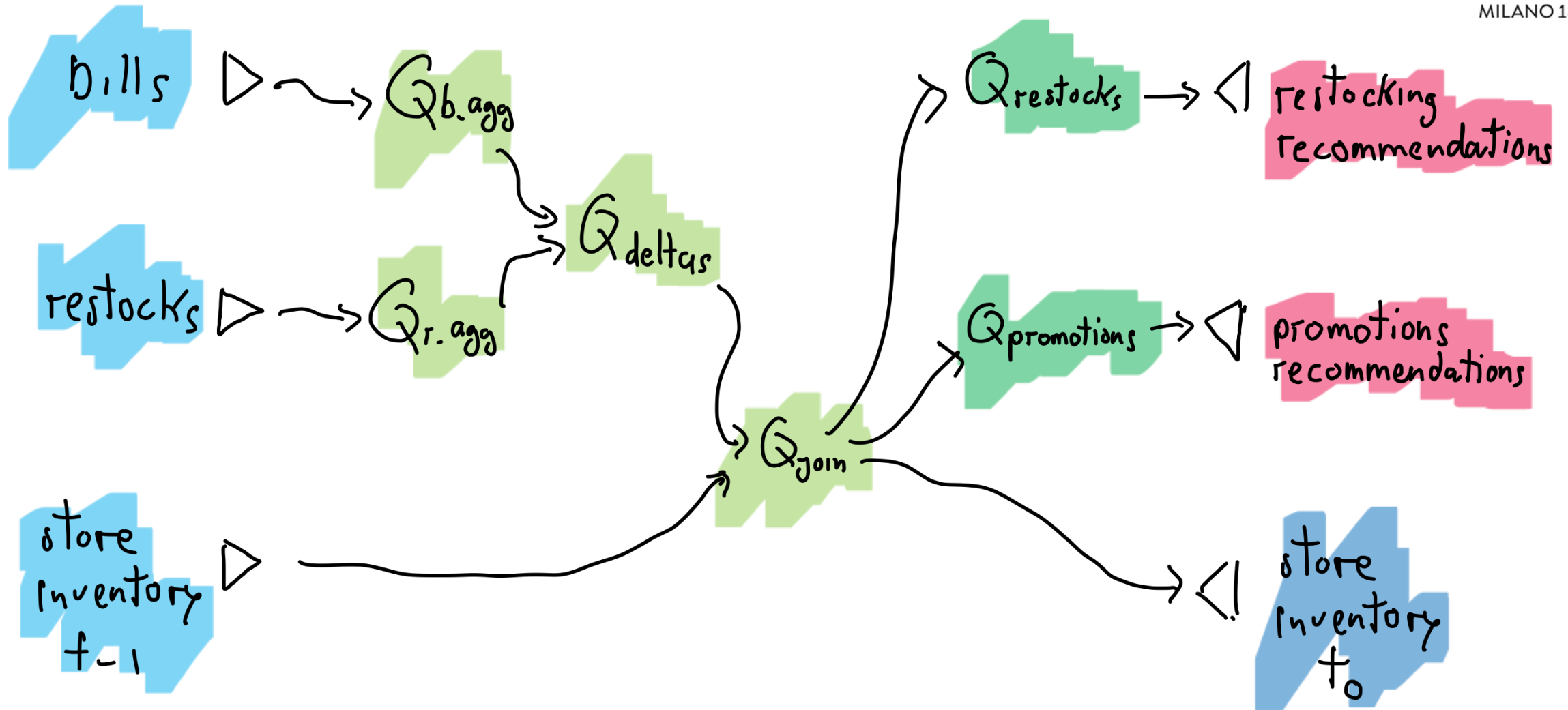




Sources

Registered Queries

Sinks





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Designing ... (at low level)



APACHE

kafka's design dimensions (template)



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- For each message
 - Key: null | simple value | structured value
 - Value: simple value | structured value
 - Typical dimension: <1KB – 1 MB
- Producers
 - Number: at need
 - Where: at need
 - How often: at need
- Brokers
 - Number: at need
 - Where: distributed | centralized | hybrid
- For each topic
 - Partition criteria: round robin | hash-based
 - Number of partitions: as many as you need
 - Number of replicas: as many as you need
 - Retention policy: time-based | size-based
 - Clean-up policy: delete old | compaction
- Consumer Groups
 - Number: at need
 - Number of consumers in the group: at need
 - Where: at need



Q: Which are Bills topic's design decisions?

- Message
 - Key:
 - Value:
 - Typical dimension:
- Producers
 - Number:
 - Where:
 - How often:
- Brokers
 - Number:
 - Where:
- Topic
 - Partition criteria:
 - Number of partitions:
 - Number of replicas:
 - Retention policy:
 - Clean-up policy:
- Consumer Groups
 - Number:
 - Number of consumers in the group:
 - Where:



A: Bills topic's design decisions (1st round)

- Message
 - Key: **null**
 - Value: **{store info, customer info, items: [{SKU, Qty, price, promotion}], totals, time-stamp}**
 - Typical dimension: **1KB-10KB**
- Producers
 - Number: **~100.000**
 - Where: **one per store | one per cash desk**
 - How often: **1/s (at peak)**
- Brokers
 - Number: at need
 - Where: distributed | centralized | hybrid
- Topic
 - Partition criteria: **round-robin**
 - Number of partitions: as many as you need
 - Number of replicas: as many as you need
 - Retention policy: time-based | size-based
 - Clean-up policy: delete old | compaction
- Consumer Groups
 - Number: at need
 - Number of consumers in the group: at need
 - Where: at need



A: Bills topic's design decisions (2nd round)

- Message

- Key: null
- Value: {store info, customer info, items: [{SKU, Qty, price}], totals}
- Typical dimension: 1KB-10KB

- Producers

- Number: 1000
- Where: one per store
- How often: 1000 times per day

- Brokers

- Number: 10
- Where: centralized | hybrid

- Topic

- Number of partitions: 1000
- Retention: 1 day
- Replicas: 3
- Size-based
- Number of consumers in the group: at need
- Where: at need



A: Bills topic's design decisions (2nd round)

- Message
 - Key: null
 - Value: {store info, customer info, items: [{SKU, Qty, price, promotion}], totals}
 - Typical dimension: 1KB-10KB
- Producers
 - Number: ~100.000
 - Where: one per store | one per cash desk
 - How often: 1/s (at peak)
- Brokers
 - Number: **10s**
 - Where: **centralized**
- Topic
 - Partition criteria: round-robin
 - Number of partitions: **10**
 - Number of replicas: **2**
 - Retention policy: time-based | size-based
 - Clean-up policy: delete old | compaction
- Consumer Groups
 - Number: at need
 - Number of consumers in the group: at need
 - Where: at need



A: Bills topic's design decisions (3rd round)

- Message
 - Key: null
 - Value: {store info, customer info, items: [{SKU, Qty, price, promotion}], totals}
 - Typical dimension: 1KB-10KB
- Producers
 - Number: ~100.000
 - Where: one per store | one per cash desk
- Brokers
 - Number: **10s**
 - Where: centralized
- Topic
 - Partition criteria: round-robin
 - Number of partitions: **10**
 - Number of replicas: **2**
 - Retention policy: **time-based**
 - Clean-up policy: **delete old**
- Consumer Groups
 - Number: **1 (Q_{b_agg})**
 - Number of consumers in the group: **10**
 - Where: **Spark Structured Streaming**



Q: Which are Restock topic's design decisions?

- Message
 - Key:
 - Value:
 - Typical dimension:
- Producers
 - Number:
 - Where:
 - How often:
- Brokers
 - Number:
 - Where:
- Topic
 - Partition criteria:
 - Number of partitions:
 - Number of replicas:
 - Retention policy:
 - Clean-up policy:
- Consumer Groups
 - Number:
 - Number of consumers in the group:
 - Where:



A: Restock topic's design decisions (1st round)

- Message
 - Key: **SKU + STORE**
 - Value: **{store info, SKU, Qty, time-stamp}**
 - Typical dimension: **1KB**
- Producers
 - Number: one per store x SKU -> $10^5 \times 10^6 \rightarrow 10^{11}$
 - Where: **each store when restocking a SKU**
 - How often: 1/day -> $\sim 1/10^5 \text{s}$
- Brokers
 - Number: at need
 - Where: distributed | centralized | hybrid
- Topic
 - Partition criteria: hash-based
 - Number of partitions: as many as you need
 - Number of replicas: as many as you need
 - Retention policy: time-based | size-based
 - Clean-up policy: delete old | compaction
- Consumer Groups
 - Number: at need
 - Number of consumers in the group: at need
 - Where: at need

$$1\text{KB} \times 10^{11} \text{ msg} / 10^5 \text{ s} = 10^6 \text{KB/s} = 1,000 \text{MB/s}$$

Thus, same as for Bills



A: Restock topic's design decisions (2nd round)

- Message
 - Key: SKU + STORE
 - Value: {store info, SKU, Qty, time-stamp}
 - Typical dimension: 1KB
- Producers
 - Number: one per store x SKU $\rightarrow 10^5 \times 10^6 \rightarrow 10^{11}$
 - Where: each store when restocking a SKU
 - How often: 1/day $\rightarrow \sim 1/10^5$ s
- Brokers
 - Number: **10s**
 - Where: centralized
- Topic
 - Partition criteria: round-robin
 - Number of partitions: **10**
 - Number of replicas: **2**
 - Retention policy: **time-based**
 - Clean-up policy: **delete old**
- Consumer Groups
 - Number: **1 (Q_{r_agg})**
 - Number of consumers in the group: **10**
 - Where: **Spark Structured Streaming**



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**And so long
and so forth
for all other topics ...**

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